

Pranav Unni

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EDUCATION

Manipal Institute of Technology (MIT) <i>B.Tech in Electronics & Computer Science – CGPA: 8.43/10</i>	Bengaluru, India <i>Aug 2024 – Expected Aug 2028</i>
Delhi Public School Ruby Park <i>Class XII (CBSE) – 84.5%</i>	Kolkata, India <i>May 2024</i>
Delhi Public School Ruby Park <i>Class X (CBSE) – 92.6%</i>	Kolkata, India <i>Jul 2022</i>

COMPETITIONS

Robotics Showcase – IEEE RAS Student Summit, Vidyashilp <i>1st Place Winner – Autonomous Underwater Vehicle (AUV)</i>	Aug 2025
- Placed 1st among all undergraduate teams at the Vidyashilp IEEE RAS Showcase; built an AUV as a first-time entry into competitive robotics, using RoboSub 2025 task specifications as the design brief. - Fine-tuned a YOLOv11m model to detect RoboSub 2025 targets (Sharks & Sawfish); built a class-balanced dataset from ~500 source images captured across varied lighting conditions, then expanded via YOLO augmentation pipeline (horizontal flips, rotation, color jitter, mixup) and annotated with bounding boxes in CVAT.ai; architected a low-latency distributed inference pipeline over ZMQ, offloading frames to an RTX 4060 GPU and streaming annotated results back to a Raspberry Pi Zero 2W for thruster control via Arduino motor shields.	

PROJECTS

bitsat-predictor 🐙 <i>FastAPI, React, PostgreSQL, Python, Pytest, UV (pkg manager)</i>	Nov 2025 – Present
- Shipped two integrated products: a Discord bot (the original project) self-hosted bare-metal on a dedicated Linux Lite machine with no desktop environment, and a web platform with a FastAPI backend on Hugging Face Spaces and a React + Vite frontend deployed on Cloudflare Pages – collectively reaching 2,000+ unique visitors. - Engineered a forecasting model on 12 years of historical exam data, applying min-max normalization and per-subject difficulty-tier score weighting to produce scenario-based rank predictions validated against real student outcomes. - Optimized the Discord bot's response latency to under 100ms via LRU caching and in-memory Matplotlib/Seaborn chart rendering on the bare-metal host; enforced CI/CD quality gates via TDD with Pytest and protected PR workflows across both the bot and web platform.	
Song2Vec 🐙 <i>Python, Numba (JIT), Ruff (linter), GitHub Actions, Signal Processing, MLOps</i>	Apr 2026 – Present
- Achieved a 200x execution speedup on the Dynamic Time Warping (DTW) algorithm via Numba JIT compilation with persistent caching, cutting per-comparison time from 2.18s to 11ms (benchmarked via line-profiler across 100 runs). - Built an end-to-end audio data engineering pipeline to quantify musical similarity, extracting features via Short-Time Fourier Transforms (STFT) and applying Sakoe-Chiba band constraints to prune the DTW search space. - Modernized MLOps infrastructure: migrated dependency management to UV, enforced PEP 8 code quality with Ruff, and configured GitHub Actions CI to block PR merges on test failures, cutting integration bugs.	
rankpredictor 🐙 <i>Bun, Prettier, Pandas, Data Engineering</i>	May 2024 – Present
- Developed a degree-3 polynomial regression model for exam rank forecasting, conducting rigorous EDA and feature engineering on crowdsourced student data to minimize RMSE and maximize R^2 to production-quality accuracy. - Published the cleaned, structured dataset to Kaggle for open-source research; the model powers a high-traffic prediction platform that has cumulatively served 50,000+ unique visitors since launch.	

CORE COMPETENCIES & CERTIFICATIONS

Languages: C, C++, Java, Python, HTML, CSS, JavaScript, TypeScript
Libraries & Frameworks: NumPy, Numba, Pandas, Matplotlib, Scikit-Learn, TensorFlow, Keras, PyTorch, Pytest, OpenCV, Streamlit, FastAPI, Astro
Developer Tools: Git, GitHub, Docker, Bun, UV, Ruff, Linux, Bash, NeoVim, VSCode
Methodologies: MLOps, Data Engineering, ETL Pipelines, RESTful API Design, CI/CD, TDD, Signal Processing, Time-Series Analysis, Object Detection, Computer Vision
Certifications: Math for ML: Multivariate Calculus (Imperial College London), TensorFlow Developer Intro & CNNs in TensorFlow (DeepLearning.AI), C for Everyone (UC Santa Cruz)